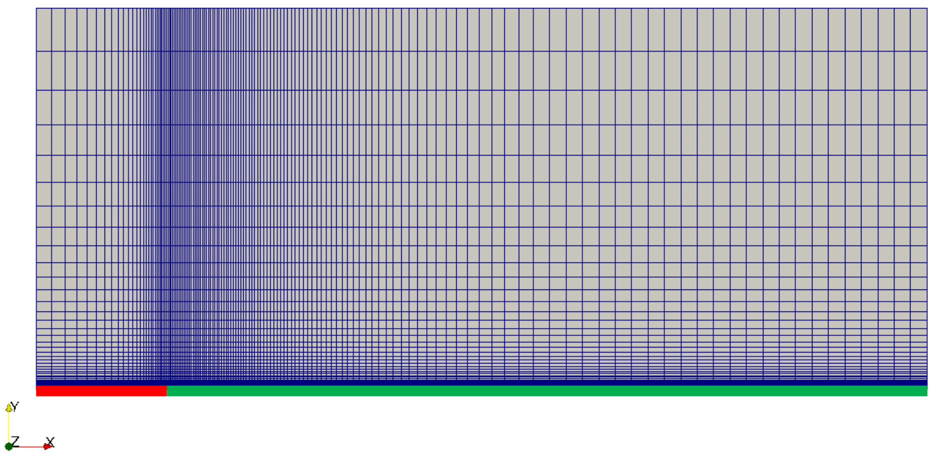
Guided tutorials

- **Guided tutorial 2.** Laminar-Turbulent flat plate.
- This case is ready to run.
- The case is located in the directory:

```
$TM/turbulence/GT2/
```

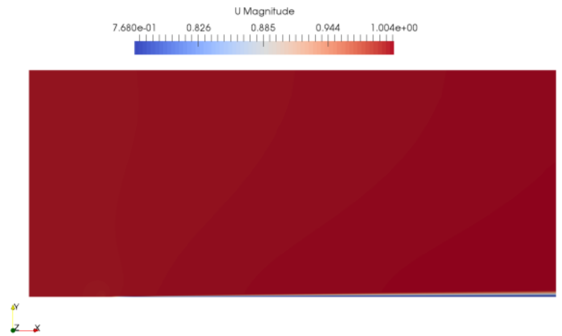
- In the case directory, you will find a few scripts with the extension .sh, namely, `run_all.sh`, `run_mesh.sh`, `run_sampling.sh`, `run_solver.sh`, and so on.
- These scripts can be used to run the case automatically by typing in the terminal, for example,
  - `$> sh run_solver`
- These scripts are human-readable, and we highly recommend you open them, get familiar with the steps, and type the commands in the terminal. In this way, you will get used with the command line interface and OpenFOAM commands.
- If you are already comfortable with OpenFOAM, run the cases automatically using these scripts.
- In the case directory, you will also find the `README.FIRST` file. In this file, you will find some additional comments.

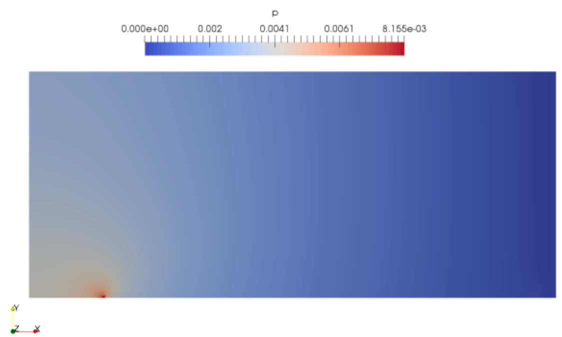
Guided tutorial 2 – Laminar-Turbulent flat plate



slip

no-slip





- In this guided tutorial we will simulate a laminar, a transitional, and a turbulent flow.
- We will use the classical 2D Zero Pressure Gradient Flat Plate validation case.
- This problem has plenty of experimental data, numerical data, and analytical correlations for validation.

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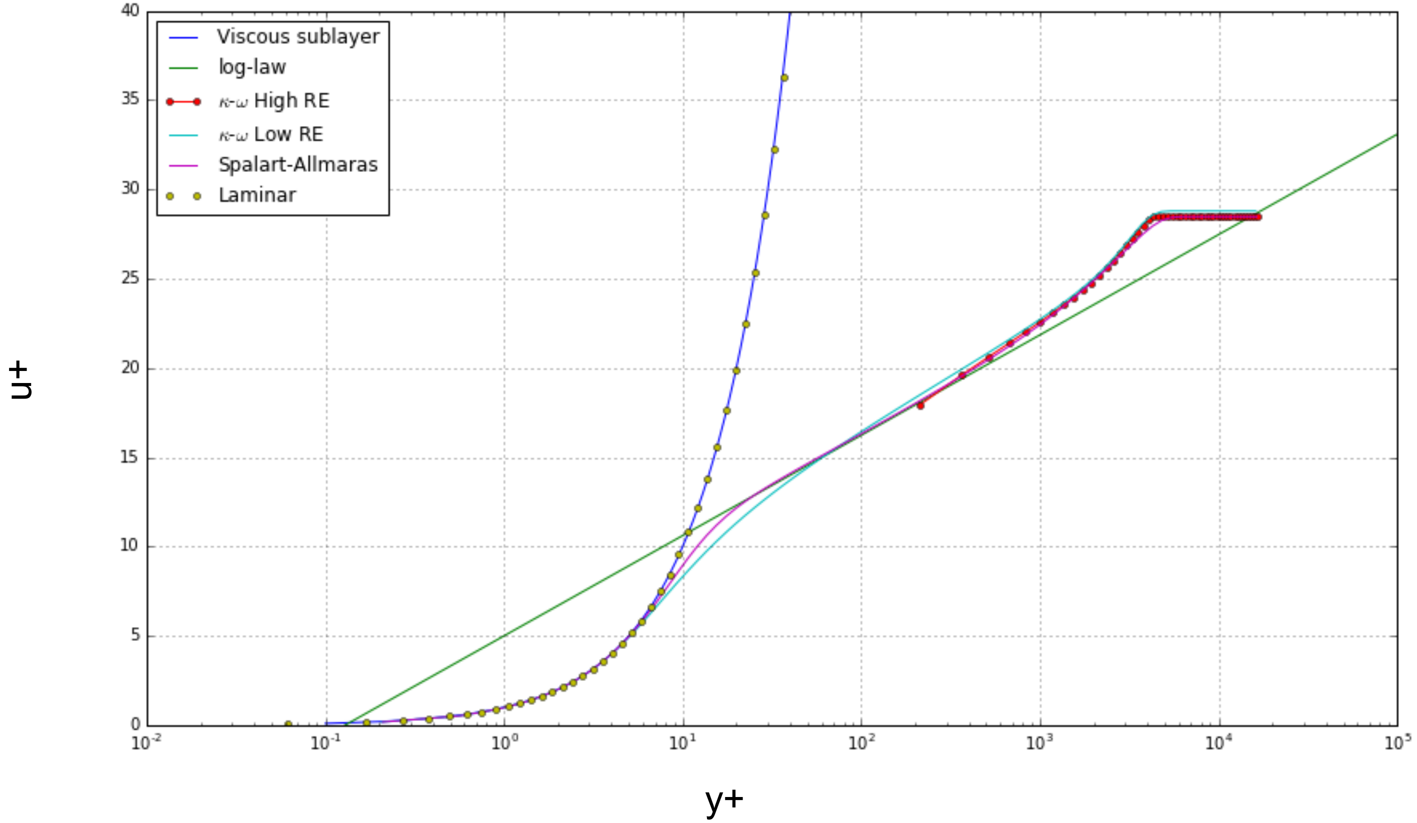

- The best way to understand the near the wall treatment and the effect of turbulence near the walls, is by reproducing the law of the wall.
- By plotting the velocity in terms of the non-dimensional variables u+ and y+, we can compare the profiles obtained from the simulations with the theoretical profiles.

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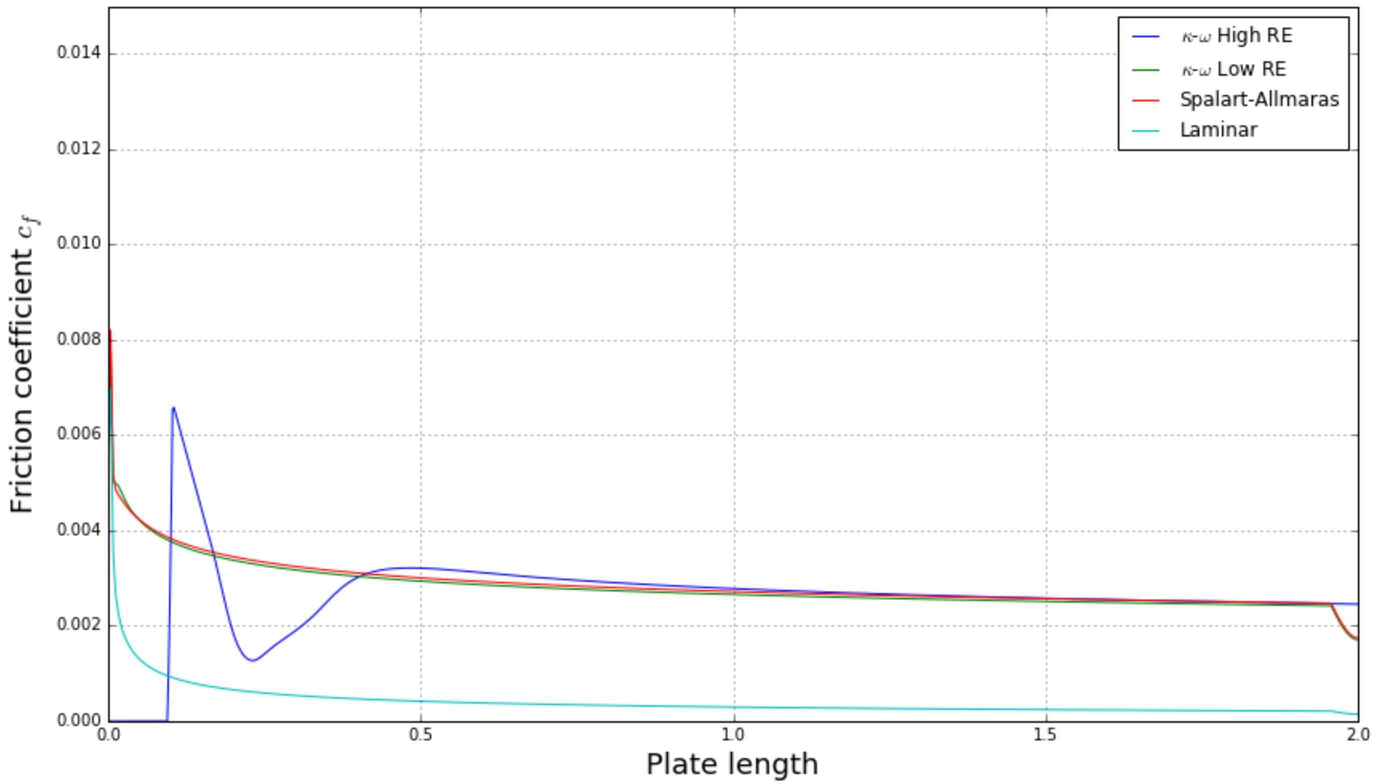

- We can also compute the friction coefficient and compare it with other results and empirical correlations.

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- In the directory **`$TM/turbulence/GT2/laminar-turbulent/samplings`** you will find a jupyter notebook that you can use to plot the non-dimensional u+ and y+ profiles of the precomputed solutions.
- In the directory **`python`** of each case, you will a jupyter notebook (a web-browser based python script), and a python script that you can use to plot the non-dimensional u+ and y+ profiles of each case.
- Remember, the u+ vs. y+ plot is kind of a universal plot. It does not matter your geometry or flow conditions, if you are resolving well the turbulent flow, you should be able to recover this profile.
- To compute this plot, you must sample the wall shear stresses.
- Then, you can compute the shear velocity, friction coefficient, and u+ and y+ values.

$$y^+ = \frac{\rho \times U_\tau \times y}{\mu} = \frac{U_\tau \times y}{\nu}$$

$$U^+ = \frac{U}{U_\tau}$$

$$U_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

$$c_f = \frac{\tau_w}{0.5 \rho U_\infty^2}$$

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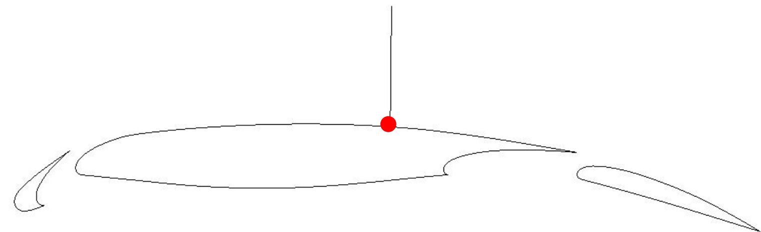
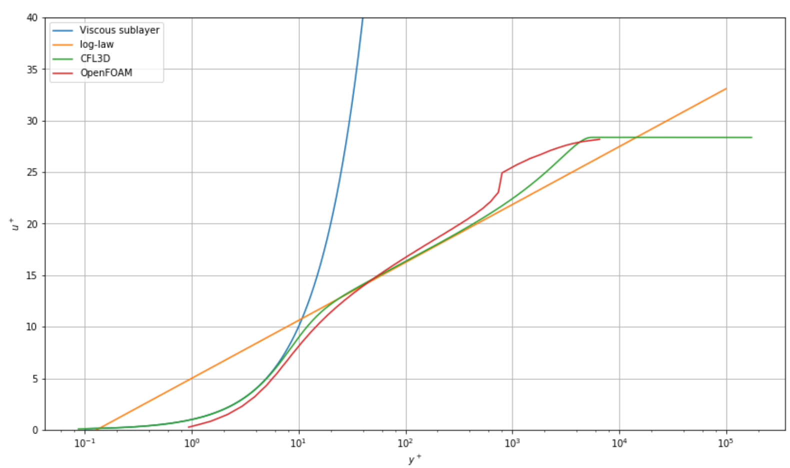
- Let us explain again how to plot the velocity in terms of the non-dimensional variables u+ and y+.
- First, pick a location where to do the sampling (avoid recirculation areas or regions with flow separation), and sample the velocity normal to the wall.
- At the same location sample the wall shear stresses and compute the magnitude.
- With the sampled values, you can compute the shear velocity, u+ and y+.
- Remember, when computing y+ you need to use the distance normal to wall y.
    - Disregarding of the location in the domain, this distance should start from zero.



Sampling location



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Guided tutorial 2 – Laminar-Turbulent flat plate

- We are going to use the `incompressibleFluid` modular solver in steady (RANS) and unsteady (LES) modes.
- This case is rather simple, but we will use it to explain many features used in OpenFOAM when dealing with turbulence, especially when dealing with near the wall treatment.
- When dealing with transitional models, we do not use wall functions.
- We will also show how to do the post-processing in order to reproduce the law of the wall. For this, we will use a jupyter notebook (or a python script) and paraview. Both approaches are valid.
- Remember, as we are introducing new closure equations for the turbulence problem, we need to define initial and boundary conditions for the new variables.
- We also need to define the discretization schemes and linear solvers to use to solve the new variables.
- It is also a good idea to setup a few functionObjects, such as: y+, minimum and maximum values, forces, time average, and online sampling.
- You will find the instructions of how to run this case in the file *`README.FIRST`* located in the case directory.